

AI-Powered Tea Quality and Moisture Prediction with Intelligent Auction Price Forecasting



Project ID: 25-26J-292



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Introduction

How does Sri Lanka's Tea Auction currently work?

- Every week Colombo hosts the largest tea auction in the world.
- Around 6.5 million kg of tea sold weekly, with 12,000+ lots.
- 8 brokers handle grading and valuation, buyers receive samples in advance.



Traditional Tea Processes & Pain Points

- Heavily depends on human expertise and judgment.
- Tasters visually examine samples
- Workers estimate moisture based on weather
- Relies on manual evaluations
- Impact: Errors reduce buyer confidence, lower quality, and affect fair pricing.

Social And Research Impact



SOCIAL

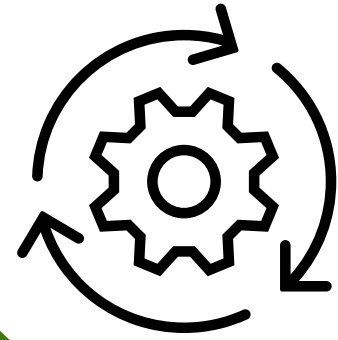
- Ensures consistent tea quality and supports fair tracking of worker performance.
- Promotes sustainable and transparent practices in the tea industry.



RESEARCH

- Real-time AI for tea grading & plucking analysis
- Predicts leaf moisture using field & environmental data
- Data-driven forecasting of tea auction prices for informed decision-making.
- Supports intelligent agriculture & workforce optimization

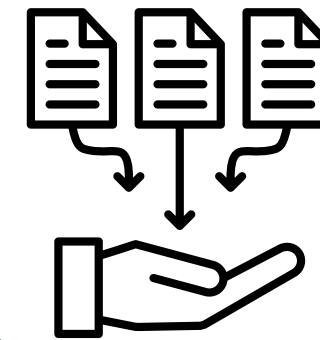
CONSTRAINTS



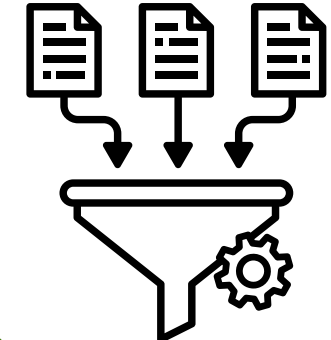
**Real-time
Processing Needs**



**Variations in leaf
appearance, lighting
and environmental
conditions.**

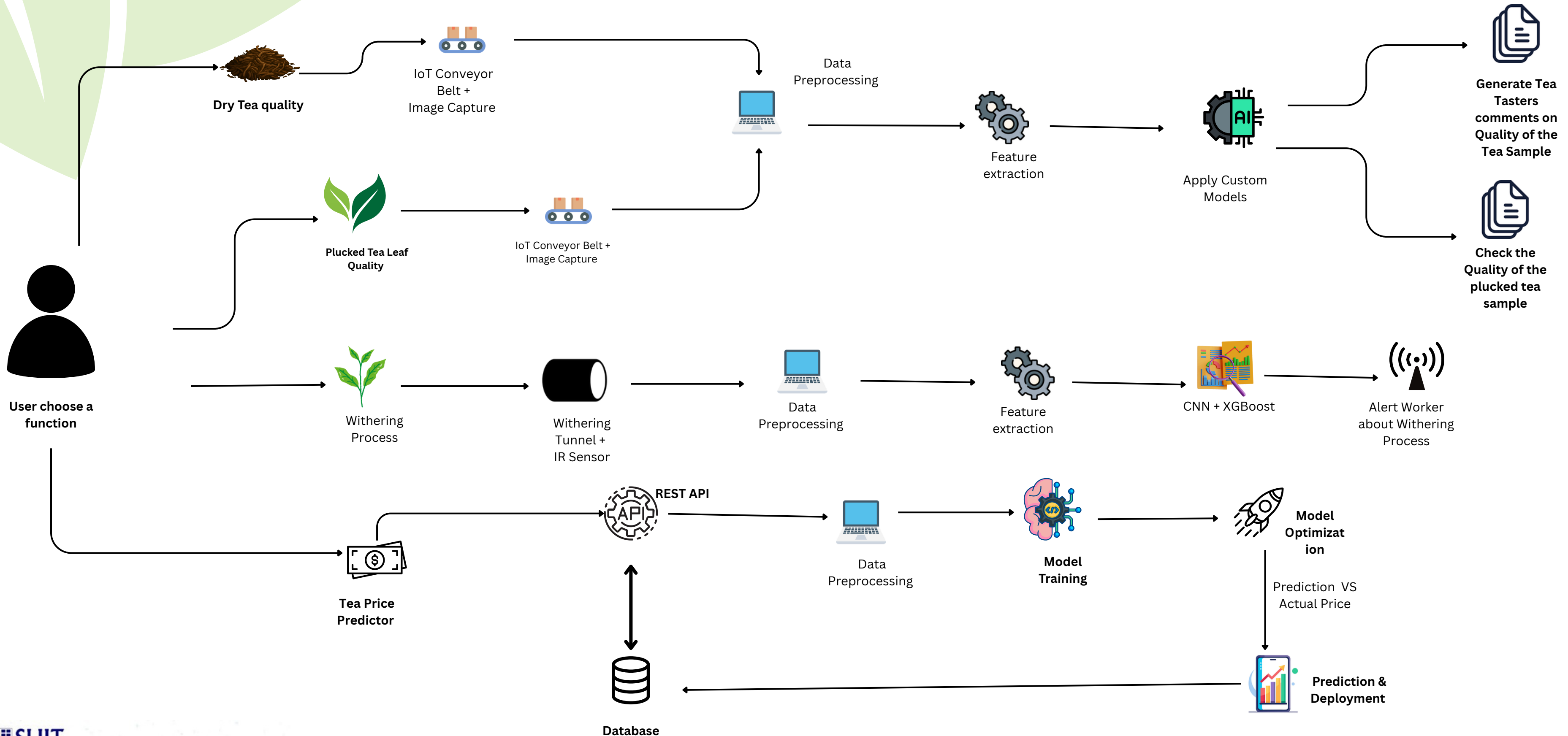


**Initial dataset covers
only a subset of tea
types, regions, and
plucking conditions.**



**Potential accuracy
drop due to system
optimization.**

System Diagram



Commercialization

Target Audience

- Tea factories (small, medium, large-scale)
- Sri Lanka Tea Board & auctioneers
- Export companies & quality certifying bodies
- International tea buyers seeking consistency

Market Space

- Global tea industry valued at USD 249B+ by 2028 (Fortune Business Insights)
- Sri Lanka exports ~300M kg annually, generating over USD 1.2B in revenue
- Growing demand for AI-driven agri-tech and quality certification solutions worldwide

Unique Selling Proposition (USP)

- End-to-end AI tea processing (grading → plucking → withering → pricing)
- Standardized, transparent, efficient auctions & exports
- Elevates Sri Lanka as a leader in smart tea technology

Previous research and its shortcomings

Symmetry detection algorithm to classify the tea grades using artificial intelligence
ScienceDirect (2020)

Shortcomings:

- Subjectivity in Visual Features
- Limited Generalization

An Automated Model for Predicting Weekly Tea Auction Prices at Colombo Tea Auction for Specific Elevations and Grades
(2024)

Shortcomings:

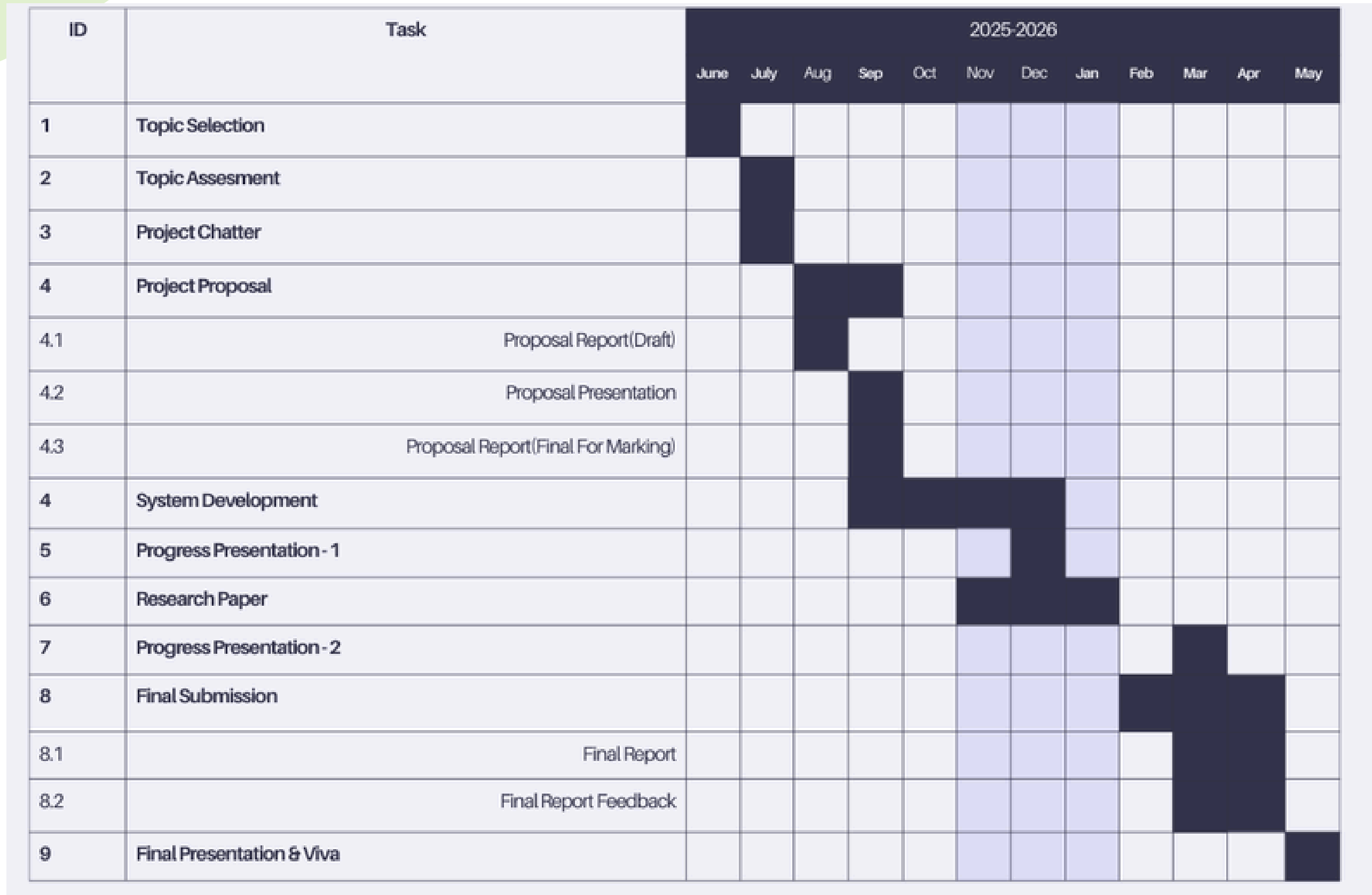
- Focused mainly on one grade (BOPF) and limited elevations.
- Predictions not fully scalable for all auction lots

Plucking Point and Posture Determination of Tea Buds Based on Deep Learning
MDPI (2025)

Shortcomings:

- Environmental Variability
- Dataset Limitations

Research Project Gantt Chart





AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features



B.Sc. (Hons) in Information Technology specializing in Information Technology

AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features



Current manual tea valuation process : visual inspection, smelling, handling leaves

AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features

Problem Definition

- Lack of objective, standardized frameworks for grading OP & BOP1 leaves.
- No validation system for human tasters, causing variability in quality assessment.



Identified Shortcomings

- Manual tea grading
- No structured training/validation for junior tasters
- Limited scalability in large auctions
- Reliance on human expertise only



Solution

AI-Powered Computer Vision System for Standardized Tea Grading



Problem Definition

Proposed Solution

Custom Hybrid Model

Instead of using only standard CNNs, develop a hybrid deep learning + handcrafted feature model tailored for tea leaf visual features (size, twist, color uniformity).

Prototype System

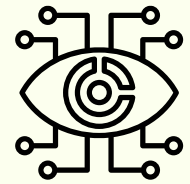
Build a portable AI-powered grading box (or camera-based device) with controlled lighting, so images are standardized and usable in real-world auctions/factories.

Continuous Learning

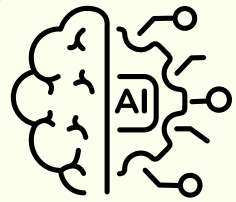
Add a mechanism where the system learns from human corrections (auction experts validating/overriding predictions), so it improves over time.

AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features

Key Domains



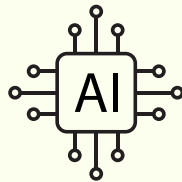
Computer Vision → Image capture & analysis of dry tea leaves



Deep Learning (CNNs) → Automated classification & feature learning



Tea Grading Expertise → Industry standards (color, twist, boldness, uniformity)



Human-AI Collaboration → AI outputs aligned with expert language

Technologies used



AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features



Evaluation / Validation

- Accuracy testing with expert-labeled OP & BOP1 tea leaf samples
- Cross-validation against senior tea tasters' scores
- Metrics: Accuracy, Precision, Recall, F1-Score, Consistency Index
- Target: $\geq 90\%$ accuracy in grade classification, $\geq 85\%$ agreement with expert tasters



Dataset Availability

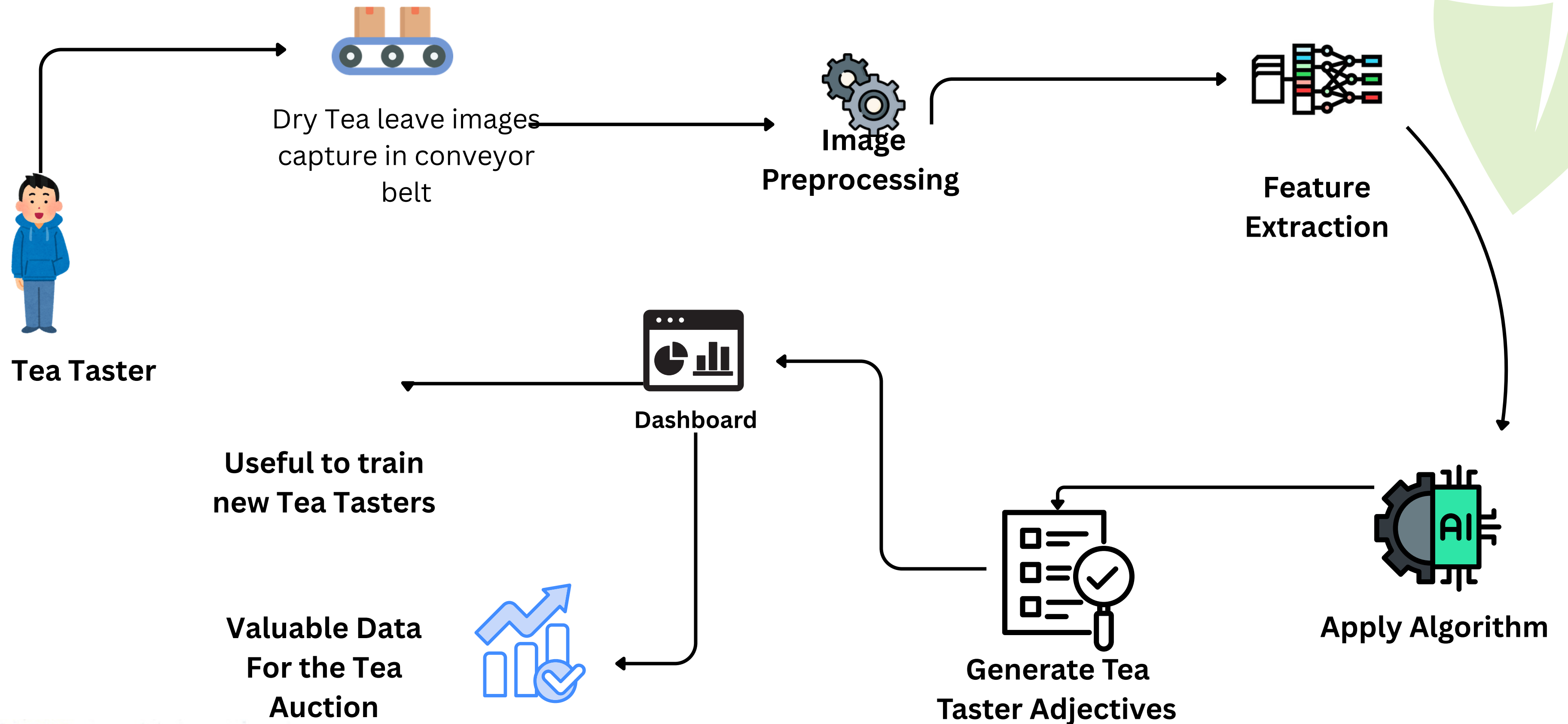
- Collect & label Sri Lankan OP & BOP1 tea leaf images under standardized conditions
- Store with metadata (grade, score, conditions)
- Open-access & reusable for research/industry



Real-World Usage

- Auctions → Consistent grading, buyer trust
- Training → Junior tasters learn with AI feedback
- Exports → Standardized grading for global branding
- Collaboration → Open dataset for research/industry

AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features





Automated Tea Leaf Inspection Using Computer Vision and Machine Learning



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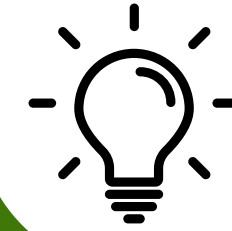
Automated Tea Leaf Inspection Using Computer Vision and Machine Learning

Problem Definition



- Plucking quality checks are manual and inconsistent.
- No real-time system to measure leaf ratio, damage, or stems.

Solution



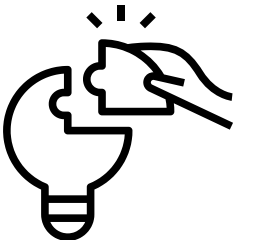
AI-Based System for Real-Time Plucking Stage Detection of Tea Leaves

Identified Shortcomings



- Primarily relies on manual inspection; no automation for grading or plucking stages.
- Large errors in assessing export-quality thresholds; cannot measure $\geq 60\%$ two leaves + bud.
- Limited detection of damages and excess stems.

Proposed Solution

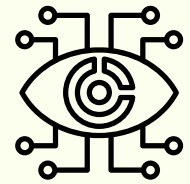


Hybrid Architecture with CNN + ML Classifier

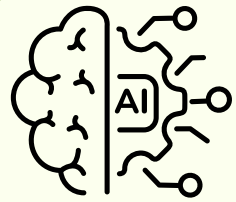
Machine Learning Model Training & Optimization. .

Automated Tea Leaf Inspection Using Computer Vision and Machine Learning

Key Domains



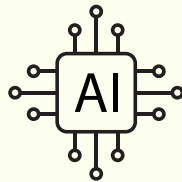
Computer Vision → Image capture & analysis of Plucked tea leaves



Deep Learning (CNNs) → Automated classification & feature learning



Tea Quality Expertise → Industry standards (two leaves and one bud formation)



Human-AI Collaboration → AI outputs aligned with expert language

Technologies used



AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features



Evaluation / Validation

- **Model Evaluation** – Train/test split, cross-validation, and performance metrics (accuracy, precision, recall, F1).
- **Error Analysis** – Confusion matrix to identify misclassifications.
- **Validation** – K-Fold and real-world testing with varied leaf conditions.
- **Optimization** – Hyperparameter tuning and feature reduction for efficiency.



Dataset Availability

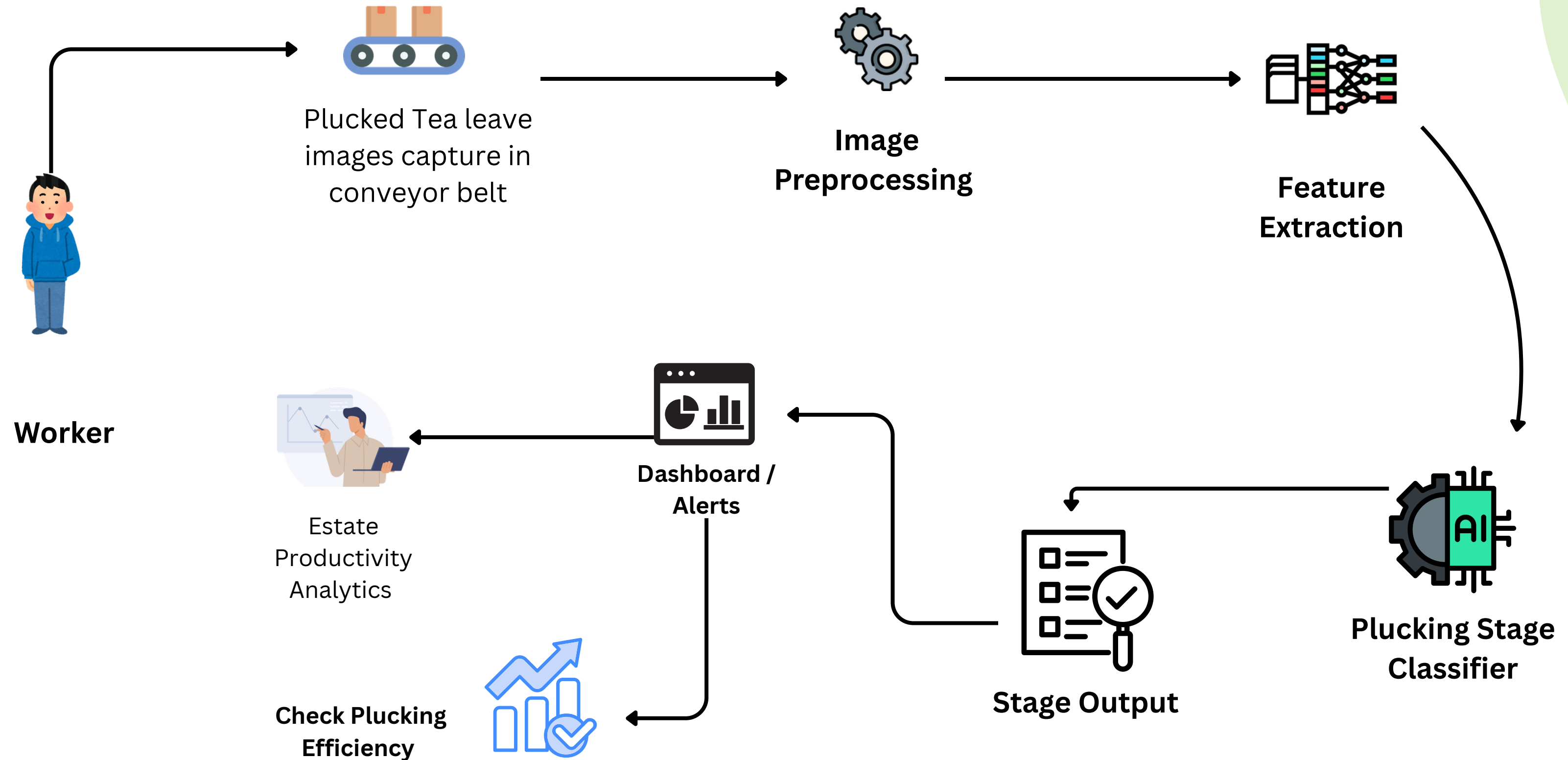
- **Collect and label tea leaf images** for different plucking stages
- **Store with clear metadata** (leaf type, plucking stage, damage, stem content)
- **Plan to make dataset open & reusable**



Real-World Usage

- **Plucking** → Consistent quality, reduced human error
- **Training** → Workers learn with AI-based feedback
- **Factories** → Faster sorting and grading process
- **Research** → Open image datasets for future studies

Automated Tea Leaf Inspection Using Computer Vision and Machine Learning





Smart Withering Monitoring and Control System

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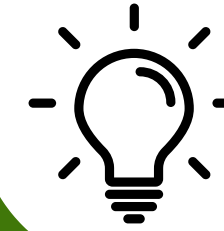
Smart Withering Monitoring and Control System

Problem Definition



- Tea withering is mostly done by workers' experience and eyesight, so it is not always correct.
- Changes in weather make it hard to know the exact withering time.

Solution



AI-Powered Vision System for Smart Tea Withering Process Monitoring

Identified Shortcomings



- Existing systems focus on only one factor (like moisture or temperature) instead of combining them.
- They still rely on workers' visual judgment and experience, which causes mistakes and inconsistent tea quality.



Problem Definition

Proposed Solution

Custom Hybrid AI Model

Combine CNNs for leaf stage recognition with time-series ML/Transformers for sensor data, using a fusion layer to predict stages and time-to-optimal.

Adaptive IoT–Edge Prototype

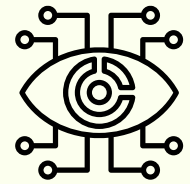
Deploy a Raspberry Pi–based system with sensors and camera, running lightweight ML locally for real-time decisions while syncing with cloud for updates.

Dynamic Withering Forecast Engine

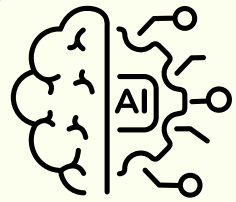
Use a regression + physics-informed ML model that adapts predictions dynamically to weather and moisture changes.

Smart Withering Monitoring and Control System

Key Domains



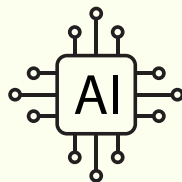
Computer Vision → Image capture & measure withering state of tea leaves



Deep Learning (CNNs) → Automated classification & feature learning

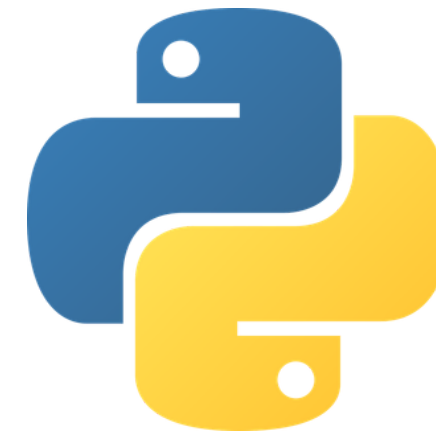


Tea Withering Expertise → Industry standards



Human-AI Collaboration → AI outputs aligned with expert language

Technologies used



TensorFlow



seaborn



NumPy

AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features



Evaluation / Validation

- **Model Evaluation:** Train/test with accuracy, precision, recall, F1.
- **Error Analysis:** Use confusion matrix to identify misclassifications.
- **Validation:** Test with real tea leaf images under varied conditions.
- **System Testing:** Perform unit and integration testing for reliability.



Dataset Availability

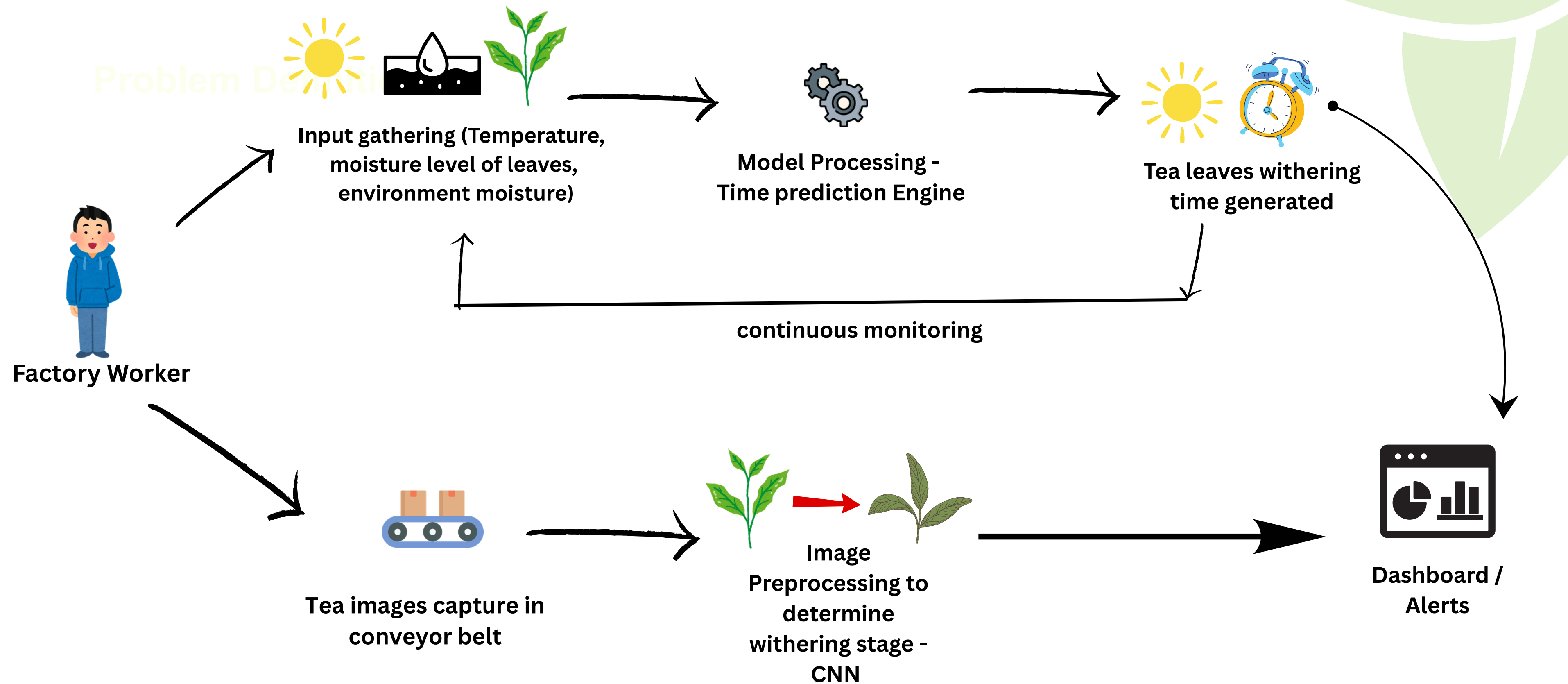
- **Collect and label tea leaf images** for different withering stages
- **Store with clear metadata** (leaf type, withering stage, damage, stem content)
- **Plan to make dataset open & reusable**



Real-World Usage

- **Moisture Control** → Ensures optimal fermentation and consistent quality
- **Efficiency** → Real-time on-field data speeds up processing
- **Quality** → Reduces human error in moisture assessment

Smart Withering Monitoring and Control System





Data-driven Tea Auction price prediction for the Colombo Tea Auction

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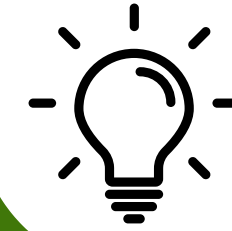
Data-driven Tea auction price prediction for the Colombo Tea auction

Problem Definition



- Current price setting relies heavily on expert judgment
- This leads to inefficiencies, subjectivity, and missed profit opportunities for stakeholders (buyers, sellers, brokers).

Solution



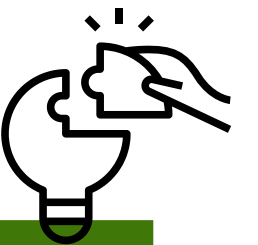
System that combines different data sources to predict tea prices in all regions with a real-time dashboard.

Identified Shortcomings



- Existing research focuses on single tea grade.
- considers limited factors (price OR quality OR weather)
- Most models consider only 4 of Sri Lanka's 7 unique agro-climatic regions

Proposed Solution



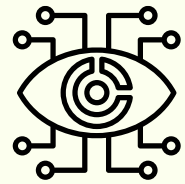
System Delivers real-time predictions through an intuitive interface.

Leverages previously ignored digital auction behavioral data.

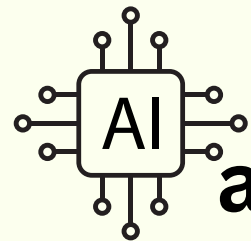
Develops a custom hybrid model tailored for tea industry data, integrating ML and deep learning to process diverse inputs.

Smart Withering Monitoring and Control System

Key Domains



Building predictive regression models (Random Forest, XGBoost, Neural Networks)



Building scalable and maintainable system architecture (APIs, Databases, UI).

Technologies used



PyTorch

research experimentation and rapid prototyping



TensorFlow

Serve infrastructure



kafka

Real-time data streaming

XGBoost

To ensure high prediction accuracy



FastAPI

Building APIs quickly and efficiently

Data-driven Tea auction price prediction for the Colombo Tea auction



Evaluation / Validation

- Prediction Accuracy (e.g., >90% accuracy for price forecasts)
-
- Latency measurement for real-time predictions



Dataset Availability

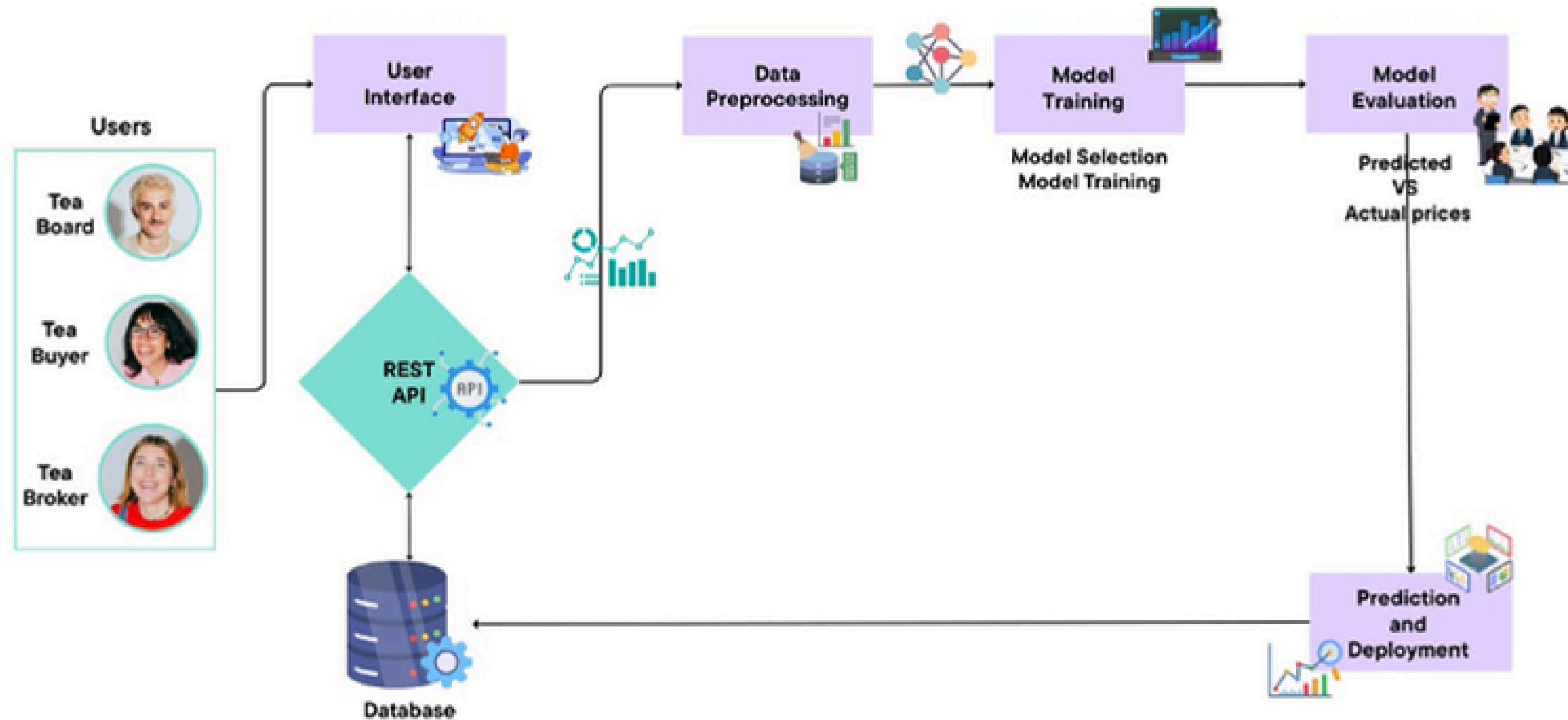
- Partnerships with major estates (e.g., Dilmah, Akbar) and the Ceylon Tea Board.
- Anonymized user session data from the digital auction platform.



Real-World Usage

- Scenario: Board bids on Nuwara Eliya tea.
- Action: System predicts price using quality, weather, and recent trends.
- Outcome: Buyer sets a confident bid, avoids overpaying.

Data-driven Tea auction price prediction for the Colombo Tea auction



Thank You



Q&A

